

Exotic Equity Derivatives Strategies : 01-Sep-15

Do Thi Hoang Ha

Macro Backdrop

Global high volatility

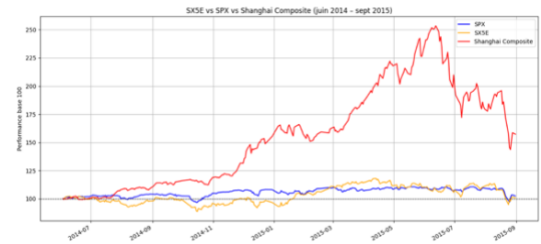
From mid-2014, the Shanghai Composite rallied over 150%, disconnected from China's decelerating GDP growth. The bubble burst in June 2015, with the index collapsing over 40%. Global contagion followed swiftly: by late August 2015, the EuroStoxx 50 had fallen over 13% and the S&P 500 over 10% within weeks. VIX and VSTOXX spike to an intraday high of 53.3 and 45.45 on 24 August.

Low-interest rate environment

On 22 January 2015, Mario Draghi announced the ECB's first sovereign QE programme (PSPP), committing to €60bn/month in asset purchases, eliminating any meaningful yield on safe fixed income and compelling institutional investors to seek conditional returns via structured products.

Relatively low correlation between SX5E and SPX

The SPX/SX5E correlation stood at 0.52 in Q4 2015, below its historical average of 0.65, reflecting two structural divergences: the sharpest Fed/ECB policy split and contrasting economic cycles.



Market view

The market view in Q4 2015 is moderately bullish on a medium-to-long term horizon, supported by ECB QE, recovering European earnings, and stabilising Chinese markets, but with significant short-term uncertainty following the August crash.

Trade Idea

10-Year Phoenix Worst-Of on SPX, SX5E, annual observation

Autocall barrier: 100%

Coupon barrier: 80%

At the maturity: Short European PDI strike 100%, low strike 80%

Structure Choice

The Phoenix WoF captures precisely this view: it monetises the post-crash volatility spike to offer a yield-enhancement structure with higher conditional coupon, while the 80% PDI provides a buffer against moderate and temporary drawdowns. The low SPX/SX5E correlation generates a WoF premium that further enriches the coupon.

Maturity Choice

The 10Y structure benefits from a significantly larger option budget than the 5Y and 7Y structures, driven by the steeper EUR yield curve beyond 7Y. Additionally, the slightly higher implied volatility at the 10Y point of the term structure enriches the down-and-in put premium, further improving coupon financing.

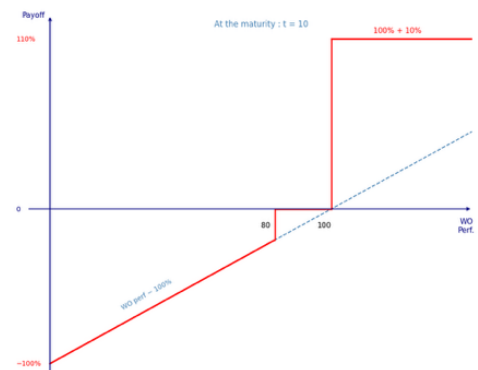
Limits and Risks

Bear market: The most direct risk of the structure is a sustained bear market, where the WoF remains below 80%, this implies no coupon is accounted, and at the maturity, investor bear the capital loss risks.

Increase in dividend yield: The SX5E is a price return index, rising dividends mechanically reduce the index level, pushing the WoF lower. With ECB expansionary monetary policy, corporate earnings can recover post-2015 and potential dividend growth of 15-20% over 3-4 years, the upside dividend risk is material.

Correlation risk: Lower correlation creates more scenarios where one index crashes while the other remains resilient. The conditional coupon compensates for this WoF premium at inception, but if correlation falls further, investors bear capital loss risk in case of single-market shocks.

Secondary market risk: The Phoenix Worst-Of is an OTC product with no secondary market, early exit is only possible via the issuer's bid-offer spread, which can widen significantly in stressed market conditions, precisely when the investor is most likely to seek liquidity.

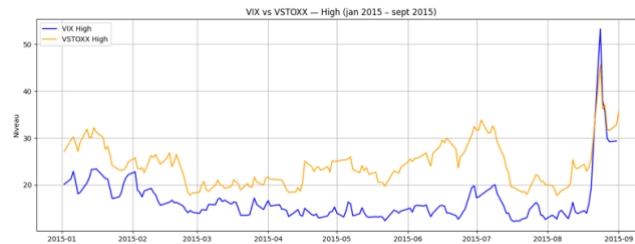


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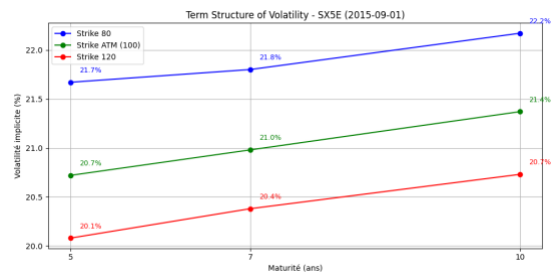
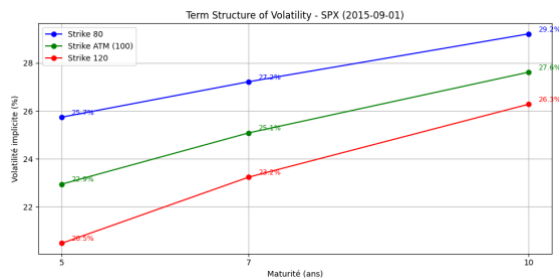
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Annexe

VIX vs VSTOXX level

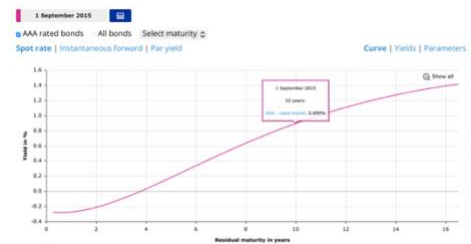


SPX vs SX5E Volatility term structure

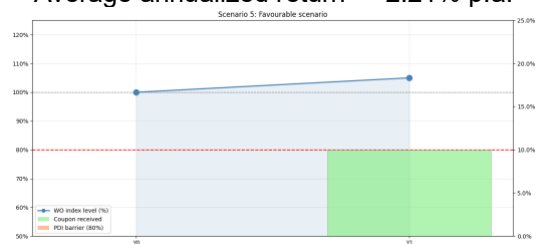
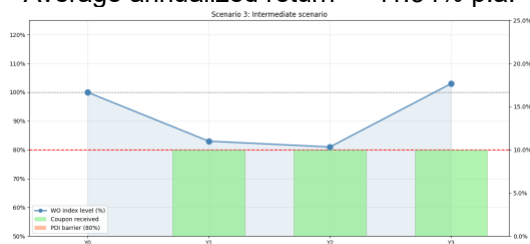
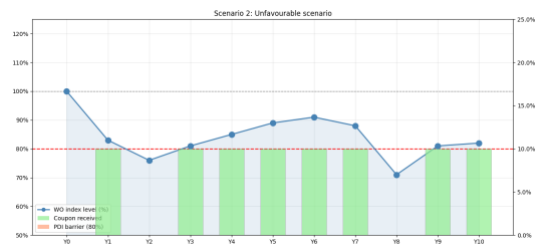
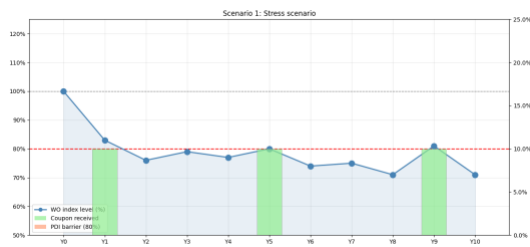


European Yield curve and option budget calculated from ZC bonds

Maturity	ZC Cost	Option Budget
5Y	99.08	0.92%
7Y	96.62	3.38%
10Y	91.44	8.56%



Scenarios Analysis



***All pricing data is in attached calculation sheet**